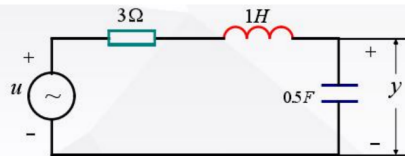


作业2 如图的RLC电路，

选 $x_1 = i, x_2 = u_c, y = u_c$

状态空间描述为



$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -3 & -1 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u$$

$$y = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

1) 设 $t_0 = 0$, $x_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, 计算系统的零输入响应

2) 设 $u(t) = 1(t)$, 计算系统的零状态响应

1)

$$x(t) = e^{At}x(0)$$

$$sI - A = \begin{pmatrix} s+3 & 1 \\ -2 & s \end{pmatrix}$$

$$(sI - A)^{-1} = \frac{1}{(s+1)(s+2)} \begin{pmatrix} s & -1 \\ 2 & s+3 \end{pmatrix} = \begin{pmatrix} \frac{-1}{s+1} + \frac{2}{s+2} & \frac{-1}{s+1} + \frac{1}{s+2} \\ \frac{2}{s+1} - \frac{2}{s+2} & \frac{1}{s+1} - \frac{1}{s+2} \end{pmatrix}$$

$$\phi(t) = e^{At} = L^{-1}[(sI - A)^{-1}] = \begin{pmatrix} -e^{-t} + 2e^{-2t} & -e^{-t} + e^{-2t} \\ 2e^{-t} - 2e^{-2t} & 2e^{-t} - e^{-2t} \end{pmatrix}$$

$$x(t) = e^{At}x(0) = \begin{pmatrix} -e^{-t} + 2e^{-2t} \\ 2e^{-t} - 2e^{-2t} \end{pmatrix}$$

$$2) x(t) = \int_0^t \phi(t-\tau)Bu(\tau)d\tau$$

$$= \int_0^t \begin{pmatrix} -e^{-(t-\tau)} + 2e^{-2(t-\tau)} & -e^{-(t-\tau)} + e^{-2(t-\tau)} \\ 2e^{-(t-\tau)} - 2e^{-2(t-\tau)} & 2e^{-(t-\tau)} - e^{-2(t-\tau)} \end{pmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} d\tau$$

$$= \begin{bmatrix} e^{-t} - e^{-2t} \\ 1 - 2e^{-t} + e^{-2t} \end{bmatrix}$$

作业：设系统的状态方程为

$$\dot{x} = \begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix} x$$

其平衡状态在坐标原点上，试判断这一状态的渐近稳定性

$$\dot{x} = \begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix} x$$

设

$$P = \begin{bmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{bmatrix}$$

由 $A^T P + P A = -Q$ 得

$$\begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix} P + P \begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

整理得：

$$\begin{cases} P_{12} + P_{21} = 1 \\ P_{11} = P_{12} + P_{22} \\ P_{11} = P_{21} + P_{22} \\ P_{12} - 2P_{22} + P_{21} = -1 \end{cases}$$

解得：

$$P = \begin{bmatrix} 3/2 & 1/2 \\ 1/2 & 1 \end{bmatrix}$$

P 为正定阵，可得该状态渐进稳定。

现代控制理论第三次作业：

作业1：分别用不同判据考察下列系统的能控性

$$(1) \quad \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$(2) \quad \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 3 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

(1)方法一：秩判据

$$U = [B \ AB \ A^2B] = \begin{bmatrix} 1 & 0 & 1 & 2 & 2 & 4 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 4 & 2 \end{bmatrix}$$
$$\text{rank}(U) = 3 = n$$

可得系统能控

方法二：PBH 判据

$$|\lambda I - A| = \begin{vmatrix} \lambda - 1 & -2 & -1 \\ 0 & \lambda - 1 & 0 \\ -1 & 0 & \lambda - 3 \end{vmatrix} = (\lambda - 1)(\lambda^2 - 4\lambda + 2) = 0$$

$$\lambda_1 = 1, \lambda_2 = 2 + \sqrt{2}, \lambda_3 = 2 - \sqrt{2}$$

$$\text{rank}(U_1) = \text{rank}[\lambda_1 I - A \ B] = \text{rank} \begin{bmatrix} 0 & -2 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ -1 & 0 & -2 & 0 & 0 \end{bmatrix} = 3 = n$$

同样可得 $\text{rank}[\lambda_2 I - A \ B] = \text{rank}[\lambda_3 I - A \ B] = 3 = n$

可得系统能控。

(2)方法一：秩判据：

$$U = [B \ AB \ A^2B] = \begin{bmatrix} 2 & 6 & 18 \\ 1 & -1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$
$$\text{rank}U = 2 < 3$$

系统不能控

方法二：PBH 判据

$$|\lambda I - A| = \begin{vmatrix} \lambda - 3 & 0 & 6 \\ 0 & \lambda + 1 & 0 \\ 0 & 0 & \lambda + 2 \end{vmatrix} = (\lambda - 3)(\lambda + 2)(\lambda + 1) = 0$$

$$\lambda_1 = 3, \quad \lambda_2 = -1, \quad \lambda_3 = -2$$

$$\text{rank}[\lambda_1 I - A \ B] = \text{rank}[\lambda_2 I - A \ B] = 3 = n$$

$$\text{rank}[\lambda_3 I - A \ B] = 2 < 3$$

综上，系统不能控

作业2：试用不同判据考察下述系统的能观性

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -2 & -4 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 & -1 \\ 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

方法一：秩判据

$$\text{rank} \begin{bmatrix} C \\ CA \\ CA^2 \end{bmatrix} = \text{rank} \begin{bmatrix} 0 & 1 & -1 \\ 1 & 2 & 1 \\ 2 & 4 & 4 \\ -2 & -3 & -1 \\ -8 & -14 & -8 \\ 2 & 2 & 0 \end{bmatrix} = 3 = n$$

系统能观

方法二：PBH 判据

$$|\lambda I - A| = \begin{vmatrix} \lambda & -1 & 0 \\ 0 & \lambda & -1 \\ 2 & 4 & \lambda + 3 \end{vmatrix} = (s+1)(s^2 + 2s + 2) = 0$$

$$\lambda_1 = -1, \quad \lambda_2 = -1 + i, \quad \lambda_3 = -1 - i$$

$$\text{rank} \begin{bmatrix} \lambda_1 I - A \\ C \end{bmatrix} = \text{rank} \begin{bmatrix} \lambda_2 I - A \\ C \end{bmatrix} = \text{rank} \begin{bmatrix} \lambda_3 I - A \\ C \end{bmatrix} = 3$$

系统能观。

作业3：试确定下列系统当p与q为何值时不能控？为何值时不能观测？

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 1 & 12 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} p \\ -1 \end{bmatrix} u$$

$$y = [q \quad 1] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$U_1 = [B \ AB] = \begin{bmatrix} p & p-12 \\ -1 & p \end{bmatrix}$$

$$\text{rank} U_1 = \text{rank} \begin{bmatrix} p & p^2 + p - 12 \\ -1 & 0 \end{bmatrix}$$

当且仅当 $p^2 + p - 12 = 0$ ，即 $p = -4$ 或 $p = 3$ 时， $\text{rank} U_1 = 1 < 2$ ，系统不可控。

$$U_2 = \begin{bmatrix} C \\ CA \end{bmatrix} = \begin{bmatrix} q & 1 \\ q+1 & 12q \end{bmatrix}$$

$$\text{rank} U_2 = \text{rank} \begin{bmatrix} q & 1 \\ -12q^2 + q + 1 & 0 \end{bmatrix}$$

当且仅当 $-12q^2 + q + 1 = 0$ ，即 $q = \frac{1}{3}$ 或 $q = -\frac{1}{4}$ 时， $\text{rank} U_2 = 1 < 2$ ，系统不能观。

作业4: 设 Σ_1 和 Σ_2 是两个能控且能观的系统

$$\begin{aligned}\Sigma_1: A_1 &= \begin{bmatrix} 0 & 1 \\ -3 & -4 \end{bmatrix}, & b_1 &= \begin{bmatrix} 0 \\ 1 \end{bmatrix}, & c_1 &= [2 \quad 1] \\ \Sigma_2: A_2 &= -2, & b_2 &= 1, & c_2 &= 1\end{aligned}$$

试分析 Σ_1 和 Σ_2 所组成的串联系统的能控性和能观性，并写出其传递函数

两系统串联后

$$A = \begin{bmatrix} A_1 & 0 \\ b_2 c_1 & A_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ -3 & -4 & 0 \\ 2 & 1 & -2 \end{bmatrix}, B = \begin{bmatrix} b_1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, C = [0 \quad c_2] = [0 \quad 0 \quad 1]$$

$$U_1 = [B \quad AB \quad A^2 B] = \begin{bmatrix} 0 & 1 & -4 \\ 1 & -4 & 9 \\ 0 & 1 & 4 \end{bmatrix}, \text{rank} U_1 = 2, \text{系统能控}$$

$$U_2 = \begin{bmatrix} C \\ CA \\ CA^2 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 2 & 1 & -2 \\ -7 & -4 & 4 \end{bmatrix}, \text{rank} U_2 = 3, \text{系统能观}$$

$$G(s) = C[sI - A]^{-1}B = \frac{1}{(s+1)(s+3)}$$